Emergency Triage, Red Flags, Vascular / Myelopathy Screen & Subjective

A · PATIENT DEMOGRAPHICS & CERVICAL SPINE PRESENTATION PROFILE — INTAKE

Patient Name _____ Age _____ Sex ☐ M ☐ F Date _____

Occupation / Posture Demand _____ Dominant Hand ☐ R ☐ L Side ☐ Central ☐ R ☐ L ☐ Bilat Vitals: BP _____ HR _____ SpO_2 _____ Temp _____

NPRS Now (circle) 0 1 2 3 4 5 6 7 8 9 10 Arm / Headache ☐ Neck only ☐ Headache ☐ Distal arm Worst _____

☐ EPA contraindications (Houghton 2010): implanted electronic devices, pregnancy, malignancy, recently irradiated tissue, active DVT, bleeding, impaired sensation — Level: Guideline

☐ Anticoagulants / coagulopathy before needling (Kearns 2019 [E]): caution — possible increased incidence or severity of haematoma and bruising

B · TRAUMA CLEARANCE (CCR / NEXUS), MYELOPATHY, CAD & UPPER CERVICAL INSTABILITY TRIAGE — TRIAGE GATE

TRAUMA IMAGING RULES & SERIOUS PATHOLOGY SCREEN — JOSPT 2017

☐ **Canadian C-Spine Rule** (Stiell 2001 JAMA) [H]: 3 questions: (1) high-risk (age ≥ 65 years, dangerous mechanism, extremity paresthesias) → image; (2) low-risk (simple rear-end MVC, sitting in ED, ambulatory, delayed neck pain, no midline tenderness) → ROM check; (3) active rotation 45° L & R → no imaging. **Sn 100% (98–100%), Sp 42.5% (40–44%)**

☐ **CCR vs NEXUS** (Stiell 2003 NEJM) [H]: **Sn 99.4% vs 90.7%; Sp 45.1% vs 36.8%; radiography rates 55.9% vs 66.6% (all P < 0.001)** · **NEXUS** (Hoffman 2000 NEJM) [H]: **Sn 99.0% (98.0–99.6%); Sp 12.9%; NPV 99.8%**

☐ **Serious pathology (CPG)** (Blanpied 2017 JOSPT) (Guideline): **"Clinicians should perform assessments … to determine the potential for the presence of serious pathology (eg, infection, cancer, cardiac involvement, arterial insufficiency, upper cervical ligamentous insufficiency, unexplained cranial nerve dysfunction or fracture), and refer for consultation as indicated."**

CERVICAL ARTERY DYSFUNCTION & POSITIONAL VBI TESTING — IFOMPT

☐ **Cervical artery dysfunction** (Rushton 2023 JOSPT) (Guideline): **IFOMPT Framework: vascular pathologies of the neck are rare but must be considered; recognised mainly through history-taking and adapted physical examination**

☐ **Positional VBI tests** (Hutting 2013 · Hutting 2022) [M]: **Sn 0–57%, Sp 67–100%; "the premanipulative tests do not seem valid"; positional and craniovertebral ligament testing excluded from IFOMPT** · **CAD symptoms:** unaccustomed headache/neck pain, diplopia, dysarthria, dysphagia, drop attacks, dizziness, facial paraesthesia, ataxia, cranial nerve deficits [E]

CERVICAL MYELOPATHY CLUSTER & UPPER MOTOR NEURON SCREEN — Cook 2010

☐ **Cervical myelopathy cluster (Cook)** (Cook 2010 J Man Manip Ther) [M]: 5 items: gait deviation, + Hoffmann, inverted supinator sign, + Babinski, age > 45 y. **≤ 1 of 5 positive: LR− 0.18 (0.12–0.42) → rules out; ≥ 3 of 5: LR+ 30.9 (5.5–181.8), post-test 94%; 4 of 5: post-test 99+% (pre-test 35%)**

☐ **Myelopathy single tests** (Cook 2010) [M]: **inverted supinator sign LR+ 29.1 (5.1–171.5); Achilles hyperreflexia 7.8 (2.5–25.4); quadriceps hyperreflexia 6.9 (2.8–17.5); Hoffmann LR+ 4.9 (2.6–9.6); age > 45 y LR− 0.48 (0.26–0.85)**

☐ **Hoffmann vs cord compression** (Glaser 2001) [L]: **Sn 58%, Sp 78%** · **Myelopathy tests (SR)** (Cook 2011) [M]: **nearly all tests specific but poorly sensitive → poor screening tools alone**

UPPER CERVICAL INSTABILITY & QUEBEC TASK FORCE WAD GRADES — Instability & WAD

☐ **Sharp-Purser test (RA)** (Uitvlugt 1988 Arthritis Rheum) [L]: **Sp 96%; Sn 88% when subluxation > 4 mm**

☐ **Upper cervical instability tests (SR)** (Hutting 2013 Phys Ther) [M]: **5 studies; specificity sufficient, sensitivity variable; membrane tests best — no pooled numbers**

☐ **Whiplash (WAD) grades** (Spitzer 1995 Spine) [E]: **Quebec Task Force grades 0–IV** (0: no neck complaint/signs; I: pain/stiffness only; II: neck complaint + MSK signs; III: neck complaint + neurological signs; IV: fracture/dislocation)

TRIAGE DISPOSITION: ☐ **Cleared — proceed** ☐ Imaging required (CCR / NEXUS +) ☐ Urgent spine referral (myelopathy / progressive neuro) ☐ **Emergency referral (suspected CAD / infection / fracture)**

C · CLINICAL PATHWAY ROUTER & JOSPT CPG SUBGROUP NAVIGATION — RAPID

CLINICAL PATHWAY ROUTER — Tick route → jump to matching pages

☐ A · MOBILITY: 1 → 2 → 3 → 7 → 9 ☐ B · RADICULAR: 1 → 2 → 4 → 6 → 9 ☐ C · WAD / POST-OP: 1 → 3 → 5 → 8 → 9

D · SUBJECTIVE HISTORY, SYMPTOM BEHAVIOUR & CERVICAL / UPPER-LIMB BODY CHART — SUBJECTIVE

SYMPTOM LOCATION & JOSPT CPG SUBGROUP SIGNPOST — Clinical Signpost

☐ **Unilateral Neck / Scapular Pain & Restricted Rotation:** segmental stiffness → **Neck pain with mobility deficits (p2/p3/p7)**

☐ **Unilateral Occipital / Frontal Headache Worsened by Neck Pos.:** (+) FRT → **Neck pain with headache / CGH (p2/p3/p7)**

☐ **Post-MVC / Whiplash or Dizziness / Motor Control Deficit:** CCFT / JPE → **Movement coordination impairments / WAD (p2/p3/p8)**

☐ **Arm Pain, Paraesthesia or Dermatomal / Myotomal Deficit:** (+) Wainner / Spurling → **Neck pain with radiating pain (p2/p4/p6)**

AGGRAVATING, EASING & IRRITABILITY PROFILE — Behaviour

Aggravating _____ Easing _____

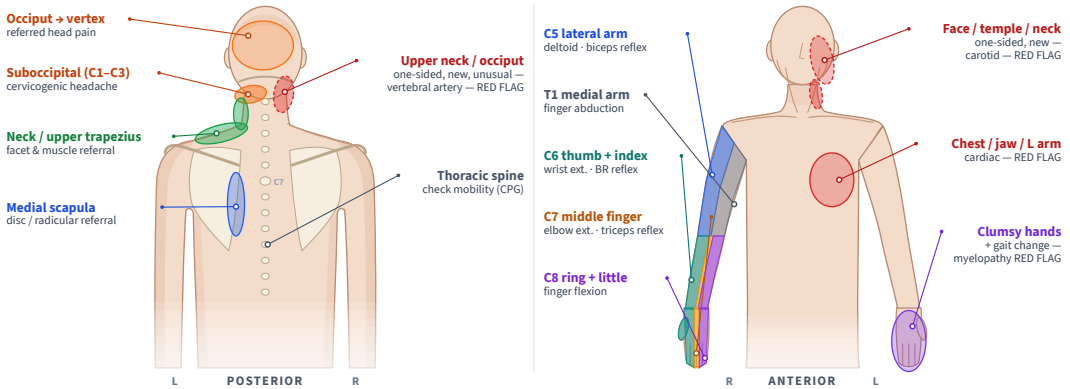
Irritability: ☐ Low ☐ Moderate ☐ High (rest/night/distal arm pain)

MECHANISM & DIURNAL PATTERN — History

Mechanism: ☐ Rear-end MVC ☐ Fall / sport trauma ☐ Sustained posture ☐ Insidious

Pattern: ☐ Morning stiffness ☐ Hand above head eases ☐ Cough / sneeze / Valsalva ☐ Constant / unremitting

BODY CHART — WHERE IS THE PAIN? — Clinical signposts [E]



Mark on the chart: shade = pain · //// = pins & needles · ×× = numbness · → = radiation · write 0–10 beside each area

Arm pain, paraesthesia or neurological signs → radiculopathy / neurodynamic screen (Wainner / ULTT A). **Gait / hand clumsiness or bilateral signs** → Cook myelopathy cluster.

SUBJECTIVE HISTORY · MECHANISM · PATIENT GOALS (PSFS)

CPG classification & prognostic stratification: classify into one of the four JOSPT 2017 CPG categories (Blanpied 2017, Guideline), screen radiculopathy (Wainner 2003 / ULTT A▸) and cervicogenic headache (FRT▸), and document baseline prognostic factors.

1 · JOSPT CPG CLASSIFICATION: MOBILITY & COORDINATION

Blanpied 2017

- ☐ **Neck pain with mobility deficits** (Blanpied 2017 JOSPT, Guideline): central and/or unilateral neck symptoms with restricted active cervical ROM, with neck pain reproduced at end range and on cervical / upper thoracic segmental mobility testing (p3/p7).
- ☐ **Neck pain with movement coordination impairments (including WAD)** (Blanpied 2017 JOSPT, Guideline): whiplash-associated or insidious neck pain with impaired deep cervical flexor activation (CCFT▸), DNF endurance▸, or sensorimotor control (JPE) (p3/p8).
- ☐ **Whiplash (WAD) grading** (Spitzer 1995 Spine) [E]: **Quebec Task Force grades 0–IV** (definitions on p1).

2 · JOSPT CPG CLASSIFICATION: HEADACHE & RADIATING PAIN

BLANPIED 2017 CPG

- ☐ **Neck pain with headaches (cervicogenic)** (Blanpied 2017 JOSPT, Guideline): unilateral/cervically triggered headache aggravated by neck movement or sustained posture, with positive flexion-rotation test▸ and upper cervical mobility restriction (p3/p7).
- ☐ **Neck pain with radiating pain (radicular)** (Blanpied 2017 JOSPT, Guideline): neck + upper-extremity radicular pain/paresthesia evaluated with ULTT A▸, Spurling▸, distraction▸, and Valsalva (p3/p4/p6).

3 · CERVICAL RADICULOPATHY CLUSTER (WAINNER 2003)

WAINNER 2003 SPINE

- ☐ **Wainner radiculopathy cluster** (Wainner 2003 Spine) [M]: **4-item cluster LR+ 30.3 (wide CI); ULTT A▸ most useful to rule out** (items: see original paper).
- ☐ **Rule-out screen▸** (Wainner 2003 / Rubinstein 2007) [M]: **ULTT A▸ Sn 0.97 (0.90–1.0), Sp 0.22 (0.12–0.33) → negative rules out** (see p4 for single-test values).

Radiculopathy Screen: ☐ ULTT A (−) rules out ☐ Wainner cluster (+) LR+ 30.3

4 · CERVICOGENIC HEADACHE — FLEXION-ROTATION TEST

OGINCE 2007 · HALL 2008

- ☐ **Flexion-rotation test (FRT) — Ogince▸** (Ogince 2007 Man Ther) [M]: **Sn 91%, Sp 90%, overall accuracy 91%**.
- ☐ **Flexion-rotation test (FRT) — Hall▸** (Hall 2008 J Manipulative Physiol Ther) [M]: **Sn 90%, Sp 88%, κ 0.85, usable by inexperienced examiners**.

FRT Status: ☐ Negative / Symmetric ☐ Positive L ☐ Positive R

5 · CLINICAL COURSE, WHIPLASH PREDICTORS & RADICULOPATHY PROGNOSIS

HUSH · WALTON · RITCHIE · WONG

- ☐ **Acute neck pain prognosis (MA)** (Hush 2011 Arch Phys Med Rehabil) [H]: **pain 64 (61–67) at onset → 35 (32–38) at 6.5 weeks → 42 (39–45) at 12 months; disability 30 → 17 by 6.5 weeks, no further improvement at 12 months**.
- ☐ **Whiplash poor-outcome predictors (MA)** (Walton 2013 JOSPT) [M]: **high baseline pain (> 5.5/10), NDI > 14.5/50, headache at inception, female sex, less than post-secondary education, WAD grade 2 or 3 (robust to publication bias)**.
- ☐ **Whiplash course + WhipPredict CPR** (Ritchie 2013 Pain) [L]: **≈ 50% fully recover, 25% persistent moderate/severe pain and disability, 25% milder disability; WhipPredict CPR PPV 71% (older age + high NDI + hyperarousal → chronic; younger + low NDI → recovery)**.
- ☐ **Radiculopathy course** (Wong 2014 Spine J) [M]: **substantial improvement within 4–6 months; complete recovery in ≈ 83% by 24–36 months; workers' compensation = poorer prognosis**.
- ☐ **WAD recovery advice (CPG)** (Blanpied 2017 JOSPT, Guideline): **"Reassurance to the patient that recovery is expected to occur within the first 2 to 3 months"**.

6 · WORKING CPG CLASSIFICATION & PROGNOSTIC PROFILE

SYNTHESIS

Primary Working Diagnosis _____ **Stage** ☐ Acute ☐ Subacute ☐ Chronic

CPG Category: ☐ Mobility deficits ☐ Movement coordination impairments (WAD) ☐ Headaches (cervicogenic) ☐ Radiating pain (radicular) ☐ Baseline pain > 5.5/10 or NDI > 14.5/50

CPG CLASSIFICATION REASONING · PROGNOSTIC RISK FACTORS · PATIENT REASSURANCE NOTES

1 · ACTIVE CERVICAL RANGE OF MOTION & FLEXION-ROTATION TEST

BLANK WRITE-IN FIELDS · NO NORMS

CERVICAL MOTION / PLANE	ACTIVE L / DEG	ACTIVE R / DEG	OVERPRESSURE	SYMPTOM REPRODUCTION & PATTERN
Cervical Flexion				☐ Local pain ☐ Peripheralises
Cervical Extension				☐ Local pain ☐ Peripheralises
Active Rotation (Left / Right)				☐ Symmetric ☐ Concordant pain
Side-Bending / Lateral Flexion (Left / Right)				☐ Symmetric ☐ Concordant pain
Flexion-Rotation Test (FRT, Left / Right) ▸ [M]				☐ Negative ☐ Positive (headache)

ROM & FRT reliability: CROM (Youdas 1992) [M]: intratester and intertester ICC generally greater than .80 • FRT (Hall 2010) [M]: mean FRT range reduced by 6° in the presence of headache, without changing test interpretation • Satpute 2019 [M]: high reliability; each test needs to be referenced to normal values.

2 · CPG EXAMINATION SET, DEEP NECK FLEXOR CONTROL & PROPRIOCEPTION

CCFT · DNF · JPE · PPT · HHD

☐ **CPG exam set** (Blanpied 2017 JOSPT, Guideline): **mobility deficits: AROM, flexion-rotation test, cervical/thoracic segmental mobility; headache: AROM, FRT, upper cervical mobility; radiating: neurodynamic, Spurling, distraction, Valsalva; coordination: cranial cervical flexion + neck flexor endurance tests; algometric pressure pain threshold.**

☐ **Craniocervical flexion test (CCFT)** ▸ (Jull 2008 J Manipulative Physiol Ther) [E]: **5 progressive pressure stages (pressure biofeedback) testing deep cervical flexor activation and endurance** • Stage / Notes: _____

☐ **Deep neck flexor endurance test:** ▸ record hold time; compare with baseline • DNF Hold Write-In: _____

Isometric Cervical Strength (HHD) · Torque & LSI Tracking [M]:

MOVEMENT	LEVER ARM (CM)	FORCE (N) INV / UNINV	TORQUE (N·M) = N × CM ÷ 100	LSI % = INV ÷ UNINV × 100
Cervical Flexion				
Cervical Extension				
Cervical Retraction				
Side-Flexion (L / R)				

Method: Versteegh 2015 [M]: intra-session ICC flexion 0.97 (0.93–0.98), extension 0.97 (0.93–0.98); MDC 2.66–4.72 kgf (trial 1–2) and 3.38–5.64 kgf (trial 2–3); flexion:extension ratio men 0.64 (0.55–0.73), women 0.55 (0.48–0.62) • Habberfield 2022 [L]: intra-rater ICC flexion 0.871, extension 0.830 • Tudini 2019 [M]: ICC .885–.974; MDC 21.1 to 47.6 N • Same tester, device, position at every visit (E).

☐ **Joint position error (JPE)** (Revel 1991 Arch Phys Med Rehabil; Treleaven 2006 Man Ther) [L]: **proprioceptive deficit in neck pain (Revel); abnormal rotation JPE: PPV 88% but Sn 60%, Sp 54% for balance problems in WAD** • JPE Write-In: _____

Segmental Mobility (C0–T4): ☐ C0–C2 hypomobile ☐ C2–C7 hypomobile ☐ C4–7 peripheralises ☐ Thoracic hypomobile **PPT Write-In:** _____

3 · UPPER-LIMB NEUROLOGICAL SCREEN (C5–T1) & UMN / MYELOPATHY ROW

DERMATOME · MYOTOME · REFLEX · UMN

ROOT	DERMATOME (LIGHT TOUCH / PINPRICK)	MYOTOME (KEY MUSCLE ACTION)	REFLEX	LEFT	RIGHT
C5	Lateral arm (regimental badge area) [E]	Shoulder abduction (Deltoid) [E]	Biceps [E]		
C6	Lateral forearm, thumb & index finger [E]	Elbow flexion & wrist extension [E]	Brachioradialis [E]		
C7	Middle finger & central palm [E]	Elbow extension & wrist flexion [E]	Triceps [E]		
C8	Ring & little fingers, medial forearm [E]	Thumb extension & finger flexion [E]	_ [E]		
T1	Medial arm & epicondylar border [E]	Finger abduction / adduction (Interossei) [E]	_ [E]		
UMN	**UMN / Myelopathy Screen (Cook 2010** [M]**):** ☐ Hoffmann ▸ ☐ Inverted supinator ▸ ☐ Babinski ☐ Gait deviation				

OBJECTIVE EXAMINATION NOTES · SEGMENTAL MOBILITY · MOTOR CONTROL & NEUROLOGICAL FINDINGS

How to read: use ULTT A ▸ (or 4 negative neurodynamic tests + negative arm squeeze) to rule out radiculopathy, and combine Spurling ▸ + axial traction ▸ + arm squeeze to rule in (Rubinstein 2007; Thoomes 2018). Underline = Atlas card · ▸ = verified video · Calc = interactive calculator.

1 · NEURODYNAMIC & PROVOCATIVE TESTS FOR CERVICAL RADICULOPATHY

RUBINSTEIN 2007 · WAINNER 2003 · THOOMES 2018

- ☐ **Provocative tests (SR)** ▸ (Rubinstein 2007 Eur Spine J) [M]: **Spurling, traction/distraction and Valsalva: low–moderate Sn, high Sp → rule in; ULTT high Sn, low Sp → negative rules out; shoulder abduction test low–moderate Sn, moderate–high Sp.**
- ☐ **Single tests — Wainner 2003 data (SR table, needle EMG/NCS reference)** (Rubinstein 2007 Eur Spine J) [M]: **ULTT A ▸: Sn 0.97 (0.90–1.0), Sp 0.22 (0.12–0.33) · Spurling A ▸: Sn 0.50 (0.27–0.73), Sp 0.86 (0.77–0.94) · distraction ▸: Sn 0.44 (0.21–0.67), Sp 0.90 (0.82–0.98) · Valsalva: Sn 0.22 (0.03–0.41), Sp 0.94 (0.88–1.0) · shoulder abduction ▸: Sn 0.17 (0.0–0.34), Sp 0.92 (0.85–0.99).**
- ☐ **Spurling & combinations (SR)** ▸ (Thoomes 2018 Spine J) [M]: **Sp 0.89–1.00; Sn 0.38–0.97; combine Spurling + axial traction + arm squeeze to rule in; 4 negative neurodynamic tests + negative arm squeeze to rule out.**

Test Battery: ☐ ULTT A ☐ Spurling ☐ Distraction ☐ Shoulder Abduction ☐ Valsalva ☐ Arm Squeeze

2 · THORACIC MANIPULATION CPR (CLELAND 2007 / 2010)

NOT VALIDATED

- ☐ **Thoracic manipulation CPR (derivation)** ▸ (Cleland 2007 Phys Ther) [L]: **≥ 3 of 6 variables: LR+ 5.5; success 54% → 86%** (items: see original paper).
- ☐ **CPR validation RCT** ▸ (Cleland 2010 Phys Ther) [M]: **CPR NOT validated; but thoracic manipulation + exercise > exercise alone for disability (all follow-ups) and pain at 1 week.**

3 · CERVICAL TRACTION CPR (RANEY 2009) Calc

NOT VALIDATED

- ☐ **Cervical traction CPR (5 items)** (Raney 2009 Eur Spine J) [L]: **peripheralisation with C4–7 mobility testing, + shoulder abduction test, age ≥ 55, + ULTT A, + distraction test.**
- ☐ **Thresholds (not validated)** (Raney 2009) [L]: **≥ 3/5: LR+ 4.81 (2.17–11.4), success 44% → 79.2%; ≥ 4/5: LR+ 23.1 (2.5–227.9) → 94.8% (not validated).**

4 · VALIDATED CERVICAL OUTCOME MEASURES (PROMS DASHBOARD)

PROMS · POPULATION-SPECIFIC MCIDS

CPG outcome measures (Blanpied 2017 JOSPT, Guideline): **"Clinicians should use validated self-report questionnaires … to identify a patient's baseline status and to monitor changes relative to pain, function, disability, and psychosocial functioning."**

INSTRUMENT & CALCULATOR	BASE	FOLLOW	VERIFIED RELIABILITY, MDC & MCID VALUES (BY POPULATION)
Neck Disability Index (NDI) Calc [M]			**Mechanical neck pain:** NDI ICC 0.50; MCID NDI 19 percentage points (Cleland 2008 Arch Phys Med Rehabil) • **Radiculopathy:** MDC NDI 13.4; MCID NDI 8.5 (Young 2010 Am J Phys Med Rehabil) • **SR:** ICC 0.50–0.98; MDC ≈ 5/50 uncomplicated neck pain, up to 10/50 radiculopathy (MacDermid 2009 JOSPT)
PSFS Calc [M]			**Radiculopathy:** MDC PSFS 3.3; MCID PSFS 2.2 (Young 2010 Am J Phys Med Rehabil)
NPRS Calc [M]			**Mechanical neck pain:** NPRS ICC 0.76; MCID NPRS 1.3 (Cleland 2008 Arch Phys Med Rehabil) • **Radiculopathy:** MDC NPRS 4.1; MCID NPRS 2.2 (Young 2010 Am J Phys Med Rehabil)
Whiplash Disability Questionnaire (WDQ) & TSK-11 (neck) [E]			**listed for tracking; no MCID used on this sheet** [E]

NEURODYNAMIC & CPR SYNTHESIS · BASELINE PROM SCORES · TARGET MCID GOALS

POST-OP RESTRICTIONS ARE SURGEON-DIRECTED [E] — follow the operating surgeon's individualised protocol for collar use, movement, and loading parameters (no fixed durations or limits).

1 · POST-OPERATIVE AIRWAY, SWALLOWING, VOICE, WOUND & NEUROLOGICAL RE-CHECK

MANDATORY EVERY VISIT

- ☐ **Post-op re-check** [E]: **swallowing / airway, voice, new arm weakness, wound, neuro change → contact surgeon.**
- ☐ **Dysphagia after ACDF (Zero-P cohort, n = 260)** (Xue 2022 Orthop Surg) [L]: **risk factors: more operated levels, longer operation time, larger dC2–7 angle change and dPSTT; of dysphagia cases 37.1% resolved by 3 months, 38.6% by 6 months, 17.1% by 12 months.**

VISIT RE-CHECK: ☐ Swallowing / airway clear ☐ Voice unchanged ☐ No new arm weakness (C5) ☐ Wound clean & dry ☐ No neuro change

2 · ACDF + PHYSIOTHERAPY VS STRUCTURED PHYSIOTHERAPY ALONE

ENGQUIST 2013 / 2017 · PEOLSSON 2013

- ☐ **ACDF + physio vs physio alone (radiculopathy, 2 y)** (Engquist 2013 Spine) [H]: **more rapid improvement in first year; "better/much better" 87% vs 62% at 12 months (P < 0.05), 81% vs 69% at 24 months (P = 0.28); NDI not different (P = 0.23); both groups improved.**
- ☐ **Same trial, physical function** (Peolsson 2013 Spine) [H]: **no difference in neck ROM, neck muscle endurance or hand function; "a structured physiotherapy program should precede a decision for ACDF".**
- ☐ **Same trial, 5–8 years** (Engquist 2017 J Neurosurg Spine) [H]: **NDI reduced 21% (surgery) vs 11% (physio), p = 0.03; neck pain −39 mm vs −19 mm; arm pain no significant difference; "better/much better" 93% vs 62%.**

3 · STRUCTURED POST-OP PHYSIOTHERAPY AFTER ACDF

WIBAULT 2018

- ☐ **Structured post-op physiotherapy after ACDF (RCT)** (Wibault 2018 J Neurosurg Spine) [M]: **only minor additional benefit at 6 months (expectation fulfilment); 61% of standard-care patients sought extra physiotherapy.**
- ☐ **ACDF — expected outcome** (Peolsson 2007 Eur Spine J) [M]: approximate 50 % probability of pain relief and little probability of functional improvement.

4 · CDA VS ACDF META-ANALYSIS

WANG 2020 (11 RCTS, 3,505 PTS)

- ☐ **CDA vs ACDF (MA, 11 RCTs, 3,505 patients)** (Wang 2020 Orthop Surg) [M]: **CDA higher overall success (2.10, 1.70–2.59); lower symptomatic adjacent-segment disease (0.46, 0.34–0.63) and secondary surgery (0.50, 0.29–0.87).**

5 · OPERATIVE PROCEDURE & SURGEON-DIRECTED PROTOCOL LOG

BLANK WRITE-IN

Procedure (ACDF / CDA / Posterior) & Operated Levels _____ **Surgery Date** _____ **Operating Surgeon** _____

Surgeon-Directed Collar / ROM / Lifting Instructions _____

6 · POST-OPERATIVE PROGRESSION CRITERIA CHECKLIST

CRITERIA TO PROGRESS

- ☐ **Surgeon clearance:** formal sign-off for active phase progression • **POST-OP RESTRICTIONS ARE SURGEON-DIRECTED** [E]
- ☐ **Surgical wound:** fully healed, dry, no signs of erythema, infection, or dehiscence [E]
- ☐ **Pain tolerance:** symptoms within acceptable pain-monitoring zones on page 8 (NPRS ≤ 5/10) [E]
- ☐ **Range of motion:** functional cervical AROM vs other side / pre-op baseline without neural tension [E]
- ☐ **Strength symmetry:** objective isometric cervical strength LSI recorded on page 3 [E]
- ☐ **Movement quality & confidence:** controlled segmental movement; patient expresses confidence to load [E]

OPERATIVE FINDINGS · SURGEON-DIRECTED RESTRICTIONS · POST-SURGICAL REHABILITATION NOTES

1 · CERVICAL COLLAR & MECHANICAL TRACTION

KUIJPER · GRAHAM · FRITZ · ROMEO

- ☐ **Collar or physio vs wait-and-see (recent radiculopathy)** (Kuijper 2009 BMJ) [H]: **collar + rest or 6 weeks physio + home exercise: extra arm-pain reduction 12 mm vs wait-and-see at 6 weeks; neck pain −17 mm (collar), −14 mm (physio)**.
- ☐ **Traction (Cochrane)** (Graham 2008 Cochrane) [M]: **no low-risk-of-bias evidence that clearly supports or refutes continuous or intermittent traction**.
- ☐ **Traction + exercise (RCT, radiculopathy)** (Fritz 2014 JOSPT) [M]: **NDI lower at 6 months with mechanical traction vs exercise alone (MD 13.3, 5.6–21.0) and vs over-door traction (8.1, 0.8–15.3); at 12 months vs exercise (9.8, 0.2–19.4)**.
- ☐ **Traction + PT (MA)** (Romeo 2018 Phys Ther) [M]: **mechanical traction: pain g −0.85 (−1.63 to −0.06) short term, −1.17 (−2.25 to −0.10) intermediate; disability g −1.05 (−1.81 to −0.28) intermediate**.

2 · LOW-LEVEL LASER & ELECTROTHERAPY

CHOW 2009 · KROELING 2013

- ☐ **Low-level laser (MA, 16 RCTs, 820 pts)** (Chow 2009 Lancet) [M]: **acute: RR 1.69 (1.22–2.33); chronic: RR 4.05 (2.74–5.98); VAS −19.86 mm (10.04–29.68); follow-up 1–22 weeks −22.07 mm (17.42–26.72)**.
- ☐ **Electrotherapy (Cochrane, 20 trials)** (Kroeling 2013 Cochrane) [L]: **very low quality; TENS inconsistent; galvanic current, iontophoresis, EMS no better than placebo**.

3 · JOSPT 2017 CPG & OPTIMA (NAD GRADES I–III) PASSIVE MODALITY RECOMMENDATIONS

BLANPIED 2017 · CÔTÉ 2016

- ☐ **JOSPT CPG — radiating pain** Blanpied 2017 (Guideline): acute: **"may provide mobilizing and stabilizing exercises, laser, and short-term use of a cervical collar"**; chronic: **"should provide mechanical intermittent cervical traction, combined with other interventions such as stretching and strengthening exercise plus cervical and thoracic mobilization/manipulation"**.
- ☐ **JOSPT CPG — chronic mobility deficits / WAD** (Blanpied 2017 JOSPT, Guideline): chronic mobility deficits: **"Dry needling, laser, or intermittent mechanical/manual traction"** within a multimodal approach; WAD (low risk) and chronic WAD: may provide TENS.
- ☐ **OPTIMa (NAD grades I–III)** Côté 2016 (Guideline): grades I–II ≤ 3 months: **"should not offer … cervical collar, electroacupuncture, electrotherapy, or clinic-based heat"**; grades I–II > 3 months: may consider low-level laser; should not offer electrotherapy, shortwave diathermy, clinic-based heat; grade III ≤ 3 months: should not offer cervical collar, low-level laser or traction; grade III > 3 months: no collar.

4 · CERVICAL TRACTION & ELECTROPHYSICAL MODALITY PARAMETER LOG (BLANK WRITE-IN FIELDS)

PARAMETER LOG

RX	MODALITY / INTERVENTION	POSITION / TARGET SEGMENT	FORCE / PARAMETERS	DURATION / HOLD	RESPONSE & PAIRED EXERCISE
☐	Mechanical intermittent cervical traction				
☐	Manual cervical traction + exercise				
☐	Low-level laser therapy (LLLT)				
☐	Short-term collar / TENS (selected subgroup)				
☐					

Traction: time-limited trial of an adjunct TIME-LIMITED TRIAL

Raney 2009 CPR (derivation only — not validated) [L]: ≥ 3 of 5: LR+ 4.81 (2.17–11.4), success 44 → 79.2 %; ≥ 4 of 5: LR+ 23.1, 94.8 %; rule needs validation • **Young 2009** [M]: no significant additional benefit • **Graham 2008 Cochrane** [M]: no evidence that clearly supports or refutes traction.

Trial Start Date: _____ Agreed Review Date: _____ ☐ Stop immediately if no objective change

TRACTION POSITIONING · MULTIMODAL EXERCISE PAIRING NOTES

1 · JOSPT CPG & MANIPULATION VS MOBILISATION

BLANPIED 2017 · GROSS 2015

- ☐ **JOSPT CPG — mobility deficits** Blanpied 2017 (Guideline): acute: **"should provide thoracic manipulation, a program of neck ROM exercises, and scapulothoracic and upper extremity strengthening"**; **"may provide cervical manipulation and/or mobilization"**.
- ☐ **JOSPT CPG — chronic headache** (Blanpied 2017 JOSPT, Guideline): **"cervical or cervicothoracic manipulation or mobilizations combined with shoulder girdle and neck stretching, strengthening, and endurance exercise"**.
- ☐ **Manipulation vs mobilisation (Cochrane)** (Gross 2015 Cochrane) [M]: **similar results for every outcome; because of rare but serious adverse events, research should focus on mobilisation**.

2 · THORACIC MANIPULATION EVIDENCE

MASARACCHIO 2019 · CROSS 2011

- ☐ **Thoracic manipulation (MA)** (Masaracchio 2019 PLoS One) [M]: **vs mobilisation: pain MD −13.63 (−21.79 to −5.46), disability −9.93 (−14.38 to −5.48); vs standard care: pain −13.21, disability −11.36 short term; vs cervical manipulation: no difference**.
- ☐ **Thoracic manipulation (SR)** (Cross 2011 JOSPT) [L]: **short-term improvement in acute/subacute mechanical neck pain; literature weak**.

3 · CERVICOGENIC HEADACHE TRIALS (C1–C2 SELF-SNAG & JULL 2002)

HALL 2007 · JULL 2002

- ☐ **C1–C2 self-SNAG (cervicogenic headache RCT)** (Hall 2007 JOSPT) [M]: **FRT range +15° vs +5° placebo; headache index 31 vs 51 at 4 weeks, 24 vs 44 at 12 months; overall reduction 54%**.
- ☐ **Manipulation + exercise (cervicogenic headache RCT)** (Jull 2002 Spine) [H]: **both reduced headache frequency and intensity, maintained at 12 months; combined not significantly superior (10% more relieved)**.

4 · DRY NEEDLING VS PHYSICAL MODALITIES

CHEN 2025 MA

- ☐ **Dry needling (MA vs physical modalities)** (Chen 2025 J Bodyw Mov Ther) [L]: **no difference in pain (VAS MD −0.13 cm) or NDI**.
- ☐ **Manual therapy delivery log:** Segment: _____ Technique: _____ Pre/Post FRT ▸ / AROM: _____

5 · ADVERSE EVENTS, VASCULAR SAFETY SCREEN & IFOMPT FRAMEWORK

CARNES 2010 · SWAIT & FINCH 2017 · IFOMPT

- ☐ **Adverse events (SR)** (Carnes 2010 Man Ther) [M]: **minor/moderate transient events ≈ 41% (17–68%) in cohorts, 22% (11.1–36.2%) in RCTs; major ≈ 0.13%**.
- ☐ **Serious adverse events (scoping)** (Swait & Finch 2017 Chiropr Man Therap) [L]: **1 per 2 million manipulations to 13 per 10,000 patients; association (causality unclear) with cervical arterial stroke under 45 years**.
- ☐ **Cervical artery dysfunction — IFOMPT International Framework (2023)** Rushton 2023 (Guideline): **vascular pathologies of the neck are rare but must be considered; recognised mainly through history-taking and adapted physical examination**; positional VBI testing and craniovertebral ligament testing are excluded from the IFOMPT Framework (Hutting 2022) [M].

6 · DRY NEEDLING SAFETY & ANATOMICAL DANGER ZONES

SAFETY SCREEN

- ☐ **Adverse event rates:** Brady 2014 [M]: mild AEs 19.18 %; upper risk rate for significant AEs ≤ 0.04 %; bruising 7.55 %, bleeding 4.65 % • Boyce 2020 [M]: minor AE 36.7 %; major AE < .1 % (prolonged symptom aggravation .03 %, fainting .02 %, forgotten needles .01 %; major AEs most often with gluteal, lumbar paraspinal, suboccipital and upper trapezius needling)
- ☐ **Serious risks & screening:** Halle 2016 [E]: pneumothorax, cardiac tamponade, haematoma, infection, CNS injury; filament needles 32–36 gauge (.25–.20 mm) • Kearns 2019 [E]: anticoagulants / coagulopathy — caution — possible increased incidence or severity of haematoma and bruising
- ☐ **Cervical / scapular danger zones:** Kearns 2022 [L]: upper trapezius — a 30 mm needle never reached the pleural space; consider needles < 40 mm • Valera-Calero 2021 [M]: rhomboid major — insert a maximum of 19 mm to reach the deep limit of rhomboid major; sex, BMI and thorax circumference explain 69.7 % of pleura depth • Okçu 2026 [L]: parascapular region — women: shorter skin-to-pleura distance; depth increases with BMI

CONSENT OBTAINED FOR: ☐ Cervical mobilisation / SNAG ☐ Thoracic manipulation ☐ Cervical manipulation ☐ Dry needling | ☐ Risks & alternatives discussed ☐ Patient may decline

Signature _____

MANUAL THERAPY CLINICAL REASONING · IFOMPT VASCULAR SCREEN · IMMEDIATE RE-TEST NOTES

1 · EXERCISE FOR NECK PAIN & HEADACHE

GROSS 2015 COCHRANE · JULL 2002

- ☐ **Exercise for neck pain (Cochrane, 27 trials)** (Gross 2015 Cochrane) [M]: **acute: no evidence; chronic: cervico-scapulothoracic + upper-limb strengthening — pain SMD −0.71 (−1.33 to −0.10); combined strengthening + stretching — function SMD −0.45 (−0.72 to −0.18)**.
- ☐ **Manipulation + exercise (cervicogenic headache RCT)** (Jull 2002 Spine) [H]: **both reduced headache frequency and intensity, maintained at 12 months; combined not significantly superior (10% more relieved)**.

2 · WHIPLASH TRIALS (MINT & PROMISE)

LAMB 2013 · MICHALEFF 2014

- ☐ **MINT (acute whiplash, n = 3,851 + 599)** (Lamb 2013 Lancet) [H]: **active ED management: NDI difference 0.5 (−1.5 to 2.5) at 12 months — no benefit; physio package vs single advice: NDI −3.7 (−6.1 to −1.3) at 4 months only; not cost-effective**.
- ☐ **PROMISE (chronic whiplash, n = 172)** (Michaleff 2014 Lancet) [H]: **12-week exercise programme = advice: pain difference 0.0 (−0.7 to 0.7) at 14 weeks, −0.1 (−0.8 to 0.6) at 12 months**.

3 · GUIDELINE EDUCATION & CRANIOCERVICAL / DEEP NECK FLEXOR PROGRESSION

BLANPIED 2017 · CÔTÉ 2016 · JULL 2008

- ☐ **WAD acute (CPG)** Blanpied 2017 (Guideline): educate to return to normal activities as soon as possible, minimise collar use, postural and mobility exercises, reassure **"recovery is expected to occur within the first 2 to 3 months"**.
- ☐ **OPTIMa education** Côté 2016 (Guideline): **"educate and reassure patients about the benign and self-limited nature of the typical course of NAD grades I-III and the importance of maintaining activity and movement"**.
- ☐ **CPG — chronic radiating pain** (Blanpied 2017 JOSPT, Guideline): **"should provide education and counseling to encourage participation in occupational and exercise activities"**.
- ☐ **CCFT / DNF training progression** ▸ [E]: stage-by-stage low-load craniocervical flexion motor control retraining without superficial sternocleidomastoid/scalene substitution, progressing to deep neck flexor endurance ▸ in upright postures and cervico-scapulothoracic + upper-limb strengthening.

4 · PAIN-MONITORING MODEL & ACTIVITY MODIFICATION

TRAFFIC LIGHT

- ☐ **Pain-monitoring model:** Sprague 2021 [E]: **pain should not exceed 5/10 on the NPRS during or immediately after activity and should return to pre-activity levels by the following morning** • Silbernagel 2007 [M]: continued tendon loading with a pain-monitoring model showed no negative effects
- ☐ **Origin of the zones:** Cavaggion 2024 [E]: low (0–2), acceptable (2–5) and high (5–10) risk zones were based on clinical experience

Low 0–2 (Safe) Acceptable 2–5 (Monitor) High 5–10 (Excessive)

Activity Guidance (E):

AVOID	MODIFY	OK

5 · 12-WEEK CERVICAL EXERCISE & RETURN-TO-ACTIVITY PROGRESSION LOG

12-WEEK LOG

Primary Programme Ticked: ☐ Cervico-Scapulothoracic + Upper-Limb Strengthening (Gross 2015) ☐ CCFT / DNF Motor Control & Endurance ☐ Active ROM & Self-SNAG Mobility ☐ WAD / NAD Reassurance & Early Activity

EXERCISE / REHABILITATION DOMAIN	WEEK 1 (BASELINE)	WEEK 3	WEEK 6	WEEK 9	WEEK 12 / DISCHARGE
1 · Craniocervical / DNF Motor Control					
2 · Cervico-Scapulothoracic Strengthening					
3 · Upper-Limb Strengthening & Stretching					
4 · Active Cervical ROM / Self-SNAG / JPE					
5 · Occupational & Activity Participation					

EXERCISE PRESCRIPTION DETAILS · LOAD PROGRESSION · RETURN-TO-ACTIVITY CRITERIA NOTES

Longitudinal Session-by-Session Tracking Flowsheet

Patient _____

#	DATE	NPRS	FLEX / EXT	ROT L / R	FRT / ULTT	EXERCISE & MOTOR CONTROL DELIVERED	MANUAL & TRACTION	IRRITABILITY	ADHERENCE	PATIENT SIGNATURE	INIT
1						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
2						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
3						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
4						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
5						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
6						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
7						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
8						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
9						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
10						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
11						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			
12						CCFT/DNF | Scap/UL | AROM | SNAG | Advice	Mob | Thor Manip | Tract | LLLT	Low | Mod | High			

DISCHARGE & OUTCOME MILESTONE SUMMARY (CHANGE VS VERIFIED MCID BY POPULATION)

FINAL OUTCOME

CERVICAL PROM & VERIFIED POPULATION THRESHOLD	INITIAL	DISCHARGE	Δ	MCID MET
NDI — Mechanical neck pain (Cleland 2008: ICC 0.50; MCID 19 percentage points)				☐
NDI — Radiculopathy (Young 2010: MDC 13.4; MCID 8.5)				☐
NDI — Systematic review (MacDermid 2009: MDC ≈ 5/50 uncomplicated, up to 10/50 radiculopathy)				☐
PSFS — Radiculopathy (Young 2010: MDC 3.3; MCID 2.2)				☐
NPRS — Mechanical neck pain (Cleland 2008: ICC 0.76; MCID 1.3)				☐
NPRS — Radiculopathy (Young 2010: MDC 4.1; MCID 2.2)				☐

Discharge: ☐ Goals & MCID met ☐ Self-management / Return to activity ☐ Spine / surgical consultation ☐ Non-responder

OBJECTIVE RE-TEST MILESTONES

SERIAL RE-TEST

OBJECTIVE MEASURE	BASE	MID	D/C	MET
Active Cervical ROM (Flex / Ext / Rot / SB)				☐
Flexion-Rotation Test (FRT) ▸				☐
CCFT ▸ & DNF Endurance ▸				☐
Joint Position Error (JPE)				☐
ULTT A ▸ / Spurling ▸ / Distraction ▸				☐
C5–T1 Neuro & UMN Screen				☐

Session 6 checkpoint: Re-score the main PROM against the MCID / MDC already on the sheet. If there is no change, re-assess the diagnosis, the dose or referral (E).

HOME EXERCISE MAINTENANCE · ERGONOMIC / ACTIVITY PLAN · DISCHARGE SUMMARY

Clinician / Referring follow-up _____ Signature _____ Date _____

Discharging Clinician _____ Sig. _____